Corruption-Tolerant Asynchronous Q-learning with Near-Optimal Rates

Sreejeet Maity and Aritra Mitra

Problem and method. The learner receives asynchronous transitions with Huber-corrupted, finite-variance rewards. **Robust Async-Q** uses a trimmed reward-history estimate and a dynamic threshold in each visited-pair update. **Robust Async-RAQ** adapts the threshold without known reward moments.

Assumptions and notation. The MDP has S states, A actions, and $\gamma \in (0,1)$. Clean rewards satisfy $|R(s,a)| \le \bar{R}$ and $\mathrm{Var}(r|s,a) \le \bar{\sigma}^2$; let $\widetilde{\sigma} := \max\{\bar{R}, \bar{\sigma}\}$. The ergodic behavior policy has $\lambda_{\min} := \min_{s,a} \pi(s)\mu(a|s) > 0$. The baseline uses i.i.d. stationary sampling; transitions and state-action labels remain clean. Write $e_t := \|Q_t - Q^\star\|_\infty$.

Robust update. For $x_t = (s_t, a_t)$, let $\bar{r}_t$ be its trimmed reward-history estimate. Set

$$\widetilde{r}_t = \bar{r}_t \mathbf{1}\{|\bar{r}_t| \le G_t\}, \qquad Q_{t+1}(x_t) = (1-\alpha)Q_t(x_t) + \alpha\big[\widetilde{r}_t + \gamma \max_{a'} Q_t(s_{t+1}, a')\big].$$

Other coordinates stay fixed; G_t incorporates a visitation burn-in and controls rare estimation failures.

1. Known reward moments (Theorem 2). Each bound holds w.p. at least $1-\delta$ for sufficiently large T.

$$e_T \le \frac{e_0}{T} + \mathcal{O}\left(\frac{\widetilde{\sigma}\log T}{\lambda_{\min}^{3/2}(1-\gamma)^{5/2}}\sqrt{\frac{\log(SAT/\delta)}{T}} + \frac{\bar{\sigma}\sqrt{\epsilon}}{\lambda_{\min}(1-\gamma)}\right). \tag{1}$$

2. Unknown reward moments (Theorems 4 and 6). Async-RAQ uses $m(t) = t^p$, $p \in \{1, 2, \ldots\}$.

$$\begin{aligned} \text{i.i.d.:} \quad e_T &\le \frac{e_0}{T} + \mathcal{O}\left(\frac{\widetilde{\sigma}^{1+1/(2p)}\log T}{\lambda_{\min}^{3/2}(1-\gamma)^{5/2}}\sqrt{\frac{\log(SAT/\delta)}{T}} + \frac{\bar{\sigma}\sqrt{\epsilon}}{\lambda_{\min}(1-\gamma)}\right), \\ \text{Markov:} \quad e_T &\le \frac{e_0}{T} + \mathcal{O}\left(\frac{\widetilde{\sigma}^{1+1/(2p)}\log T}{\lambda_{\min}^{3/2}(1-\gamma)^{5/2}}\sqrt{\frac{\tau\log(SAT/\delta)}{T}} + \frac{\bar{\sigma}\sqrt{\epsilon}}{\lambda_{\min}(1-\gamma)}\right). \end{aligned} \tag{2}$$

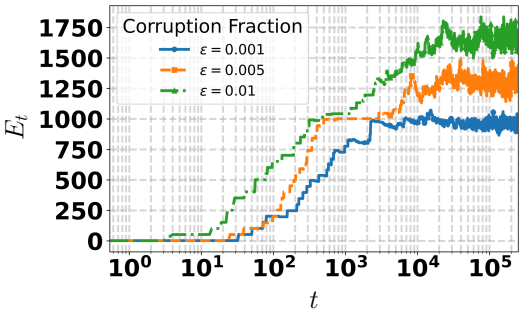
For i.i.d. data, $\alpha = \log T/[\lambda_{\min}(1-\gamma)T]$; for a stationary tuple chain with mixing time $\bar{\tau}$, subsample every $\tau = \lceil \bar{\tau}\lceil \log_2(2T/\delta)\rceil\rceil$ steps and use $\alpha_M = \tau\alpha$. Here T counts all trajectory steps.

3. Fundamental corruption limit. Even a synchronous generative observation model cannot avoid the following error: for a universal $c > 0$,

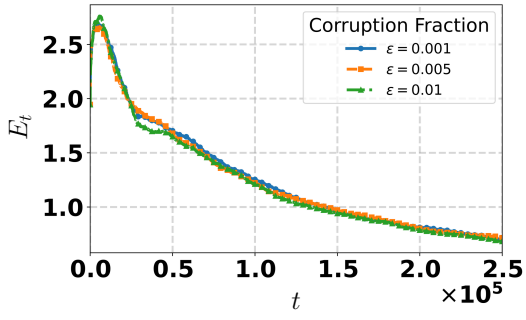
$$\inf_{\widehat{Q}_T} \sup_{\mathcal{H}(\epsilon,\bar{\sigma})} \mathbb{P}\left(\|\widehat{Q}_T - Q^\star\|_\infty \ge \frac{c\bar{\sigma}\sqrt{\epsilon}}{1-\gamma}\right) \ge \frac{1}{4}. \tag{3}$$

Thus the $\sqrt{\epsilon}$ dependence is unavoidable; the additional $\lambda_{\min}^{-1}$ factor in the upper bound remains unresolved.

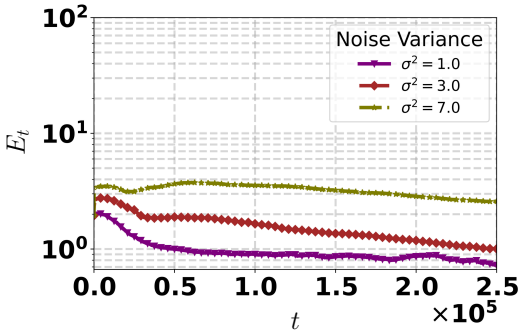
Numerical results. 100 runs; $E_t = \|Q_t - Q^\star\|_\infty$. Fig. 4(a–c): $(S, A, \gamma) = (50, 20, 0.9)$, bias -10^5; (a,b) $\sigma^2 = 0.5$, $\epsilon \in \{0.001, 0.005, 0.01\}$; (c) $\epsilon = 0.01$, $\sigma^2 \in \{1, 3, 7\}$. Fig. 1(d): $(S, A, \gamma) = (25, 10, 0.7)$, bias -10^4, $\epsilon = 0.001$, $p \in \{1, 2, 5\}$.



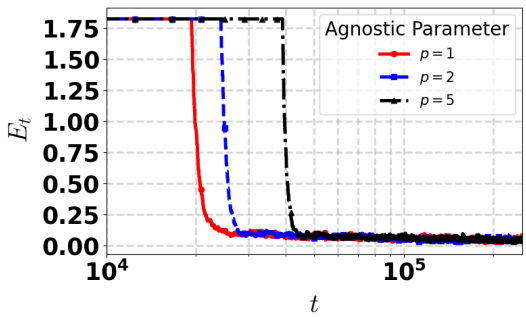
(a) Fig. 4: Vanilla-Q, varying ϵ.



(b) Fig. 4: Robust Async-Q, varying ϵ.



(c) Fig. 4: varying reward-noise variance.



(d) Fig. 1: reward-agnostic exponent p.